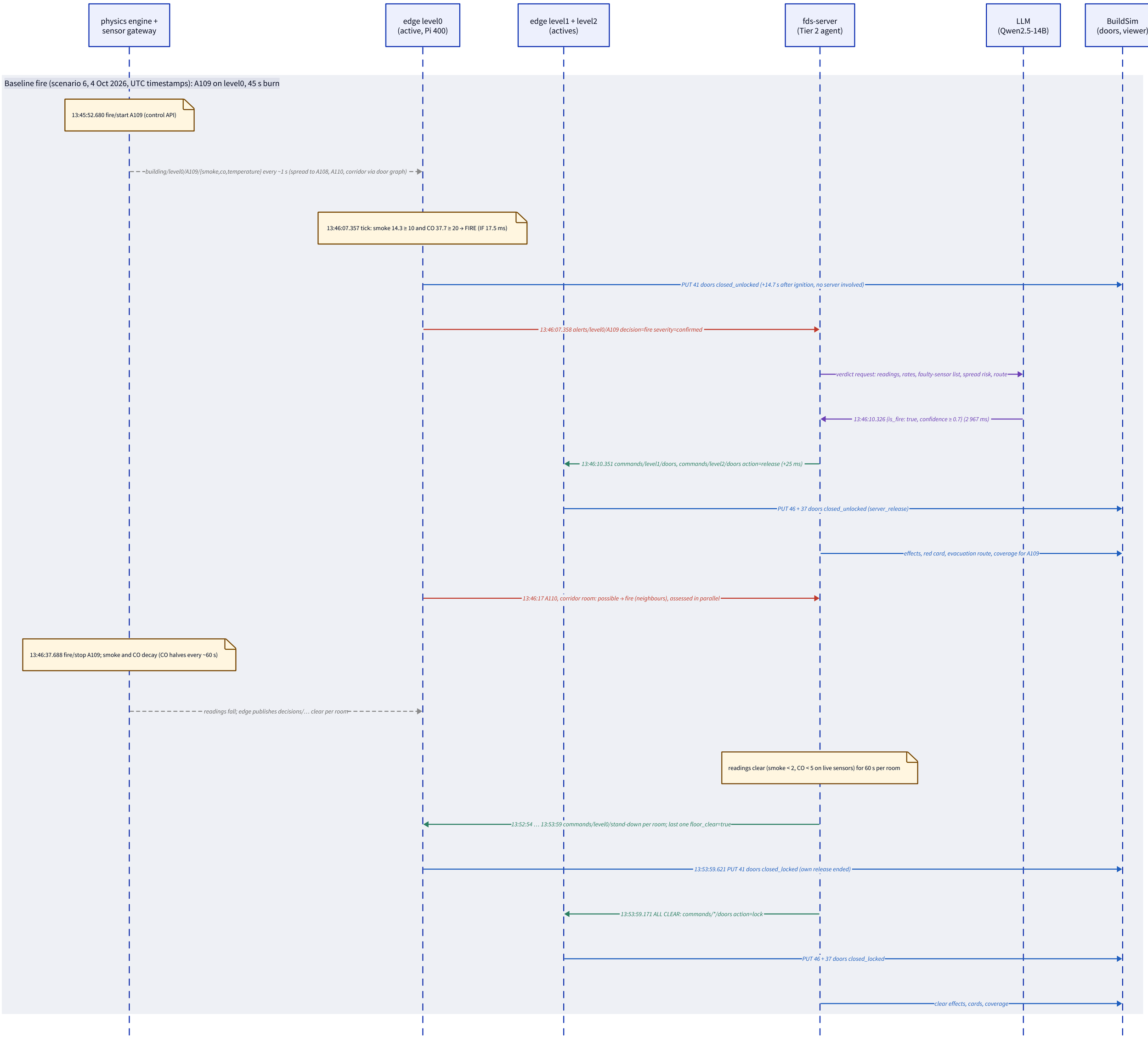

physics engine + sensor gateway
edge level0 (active, Pi 400)
edge level1 + level2 (actives)
fds-server (Tier 2 agent)
LLM (Qwen2.5-14B)
BuildSim (doors, viewer)
Baseline fire (scenario 6, 4 Oct 2026, UTC timestamps): A109 on level0, 45 s burn
13:45:52.680 fire/start A109 (control API)
building/level0/A109/{smoke,co,temperature} every ~1 s (spread to A108, A110, corridor via door graph)
13:46:07.357 tick: smoke 14.3 ≥ 10 and CO 37.7 ≥ 20 → FIRE (IF 17.5 ms)
PUT 41 doors closed_unlocked (+14.7 s after ignition, no server involved)
13:46:07.358 alerts/level0/A109 decision=fire severity=confirmed
verdict request: readings, rates, faulty-sensor list, spread risk, route
13:46:10.326 {is_fire: true, confidence ≥ 0.7} (2 967 ms)
13:46:10.351 commands/level1/doors, commands/level2/doors action=release (+25 ms)
PUT 46 + 37 doors closed_unlocked (server_release)
effects, red card, evacuation route, coverage for A109
13:46:17 A110, corridor room: possible → fire (neighbours), assessed in parallel
13:46:37.688 fire/stop A109; smoke and CO decay (CO halves every ~60 s)
readings fall; edge publishes decisions/… clear per room
readings clear (smoke < 2, CO < 5 on live sensors) for 60 s per room
13:52:54 … 13:53:59 commands/level0/stand-down per room; last one floor_clear=true
13:53:59.621 PUT 41 doors closed_locked (own release ended)
13:53:59.171 ALL CLEAR: commands/*/doors action=lock
PUT 46 + 37 doors closed_locked
clear effects, cards, coverage